Series & Binomial Theorem

ClipSAT Printable Practice Suite (clipsat.org) • Topic 10.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A particle moves along a straight line. Its velocity, v m/s, at time t seconds is given by $v = 3t - 2$. Find the displacement of the particle during the first 4 seconds of its motion.

The displacement of the particle can be found by integrating the velocity function with respect to time. The displacement s is given by: $s = \int v\,dt$ In this case, $v = 3t - 2$. $s = \int (3t - 2)\,dt$ Integrating, we get: $s = \frac{3}{2}t^2 - 2t + C$ To find the displacement during the first 4 seconds, we evaluate the definite integral from $t = 0$ to $t = 4$. We can assume $C = 0$ for calculating the change in displacement from $t = 0$. $s(4) - s(0) = \left[\frac{3}{2}t^2 - 2t\right]_0^4$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The diagram shows the graph of the function $y = f(x)$. The graph is transformed by a sequence of operations. The resulting graph, $y = g(x)$, is also shown. The function $f(x)$ is given by $f(x) = ax^2 + bx + c$, where $a, b,$ and c are constants. The vertex of the graph of $y = f(x)$ is at $(2, -1)$, and it passes through the point $(4, 3)$. The graph of $y = g(x)$ is obtained by translating the graph of $y = f(x)$ by $(-1\ 3)$ followed by a reflection in the y-axis. Find the values of $a, b,$ and c.

The vertex of the parabola $y = ax^2 + bx + c$ is at $\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$. We are given that the vertex is at $(2, -1)$. Therefore, $-\frac{b}{2a} = 2$, which implies $b = -4a$. The vertex form of the quadratic equation is $y = a(x - h)^2 + k$, where (h, k) is the vertex. So, $y = a(x - 2)^2 - 1$. The graph passes through $(4, 3)$. Substituting this point into the equation: $3 = a(4 - 2)^2 - 1 \Rightarrow 3 = a(2)^2 - 1 \Rightarrow 3 = 4a - 1 \Rightarrow 4a = 4 \Rightarrow a = 1$. Si.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A committee of 3 people is to be selected from a group of 5 people. The number of ways to form this committee is given by the combination formula ${}^nC_r = \frac{n!}{r!\,(n-r)!}$. How many different committees can be formed?

The problem requires us to find the number of combinations of selecting 3 people from a group of 5. This is a direct application of the combination formula ${}^nC_r$, where n is the total number of items to choose from, and r is the number of items to choose. In this case, $n = 5$ and $r = 3$. Using the formula: ${}^5C_3 = \frac{5!}{3!\,(5-3)!} = \frac{5!}{3!\,2!} = \frac{5 \times 4 \times 3!}{3! \times (2 \times 1)} = \frac{5 \times 4}{2} = \frac{20}{2} = 10$. Therefore, there are 10 different committees that can be forme.

© ClipSAT — clipsat.org | All Rights Reserved

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A particle of mass 3 kg is at rest on a smooth horizontal surface. A horizontal force of magnitude P N is applied to the particle, causing it to accelerate at a rate of 4 m/s^2. Find the value of P.

According to Newton's Second Law of Motion, the net force acting on an object is equal to the product of its mass and acceleration, i.e., F = ma. In this case, the mass of the particle is m = 3 kg and its acceleration is a = 4 m/s^2. The applied force P is the net force acting on the particle, as the surface is smooth and horizontal, meaning there are no other horizontal forces to consider. Therefore, P = ma = (3 kg) * (4 m/s^2) = 12 N. Thus, the value of P is 12 N.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The two-way table shows the results of a survey of 100 students on their preferred mode of transport to school and their year group. | | Car | Bus | Walk | Total | |---|---|---|---|---| | Year 1 | 15 | 20 | 10 | 45 | | Year 2 | 25 | 18 | 12 | 55 | | Total | 40 | 38 | 22 | 100 | What is the probability that a randomly selected student is a Year 1 student who prefers to travel by car?

The total number of students surveyed is 100. The number of Year 1 students who prefer to travel by car is 15. Therefore, the probability of selecting a Year 1 student who prefers to travel by car is the number of Year 1 students who prefer to travel by car divided by the total number of students. P(Year 1 and Car) = Number of (Year 1 and Car) / Total Number of Students = 15 / 100 = 0.15.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A 3 kg particle on a rough surface, $\mu = 0.4$. Horizontal force F N. (a) Find the limiting friction. (b) Acceleration when $F = 20$ N. (c) Time to travel 5 m from rest.

(a) $F_f = 0.4 \times 3 \times 9.8 = 11.76$ N. (b) $a = \frac{20 - 11.76}{3} \approx 2.75$ m s^{-2}. (c) $t = \sqrt{10/2.75} \approx 1.91$ s. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A factory tests 30 independent components, each with a fixed probability of being defective; X is the number of defective components. Is X binomially distributed?

The trials are fixed in number, independent, and each has the same probability of success, so the conditions for a binomial distribution are satisfied. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

© ClipSAT — clipsat.org | All Rights Reserved

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A survey asks 50 randomly and independently selected people the same yes/no question; X is the number who say yes. Is X binomially distributed?

The trials are fixed in number, independent, and each has the same probability of success, so the conditions for a binomial distribution are satisfied. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A machine has a useful power output of 883 W and wastes 367 W as heat. Find its efficiency as a percentage, correct to 1 decimal place.

Total input $= 883 + 367 = 1250$ W. Efficiency $= \frac{\text{useful output}}{\text{input}} \times 100\% = \frac{883}{1250} \times 100\% \approx 70.6\%$ Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

10 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A particle of mass 5 kg slides down a rough plane inclined at 30° to the horizontal, experiencing a constant friction force of 0 N. Taking $g = 9.8$ m/s^2, find its acceleration down the slope.

$W = mg = 5 \times 9.8 = 49.00$ N. Component along slope $= W\sin 30^\circ \approx 24.50$ N. Net force ≈ 24.50 N. $a = F/m \approx 4.90$ m/s^2 Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

11 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Given $X \sim N(\mu, \sigma^2)$, $P(X < 99) = 0.8$ and $P(X < 83) = 0.2$, find μ, correct to the nearest integer.

Using $z_1 = 0.8416$, $z_2 = -0.8416$ and solving $x_1 = \mu + z_1\sigma$, $x_2 = \mu + z_2\sigma$ simultaneously gives $\mu \approx 91$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

© ClipSAT — clipsat.org | All Rights Reserved

12 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A particle of mass m kg is placed on a rough inclined plane. The plane is inclined at an angle α to the horizontal, where $\tan\alpha = \frac{3}{4}$. The particle is on the point of sliding down the plane. The coefficient of static friction between the particle and the plane is μ_s. Given that the acceleration due to gravity is $g = 9.8\ \text{m/s}^2$, find the value of μ_s.

When a particle is on the point of sliding down an inclined plane, the angle of inclination α is equal to the angle of friction ϕ, where $\tan\phi = \mu_s$. Given that $\tan\alpha = \frac{3}{4}$, and the particle is on the point of sliding, we have $\alpha = \phi$. Therefore, $\mu_s = \tan\phi = \tan\alpha = \frac{3}{4}$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In an arithmetic progression, the 3rd term is 14 and the 7th term is 34. (a) Find the first term a and the common difference d. (b) Find the sum of the first 25 terms.

(a) Use $u_n = a + (n-1)d$: $u_3 = a + 2d = 14$ $u_7 = a + 6d = 34$ Subtract the first equation from the second: $4d = 20 \Rightarrow d = 5$. Substitute back: $a + 2(5) = 14 \Rightarrow a = 4$. (b) Use the sum formula $S_n = \frac{n}{2}(2a + (n-1)d)$ with $n = 25$: $S_{25} = \frac{25}{2}(2(4) + 24(5)) = \frac{25}{2}(8 + 120) = \frac{25}{2}(128) = 25 \times 64 = 1600$

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Series & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Statistics & Probability

In a manufacturing batch, 5% of components are defective. A sample of 12 components is inspected. Find the probability that at least 2 components are defective.

Let $X \sim B(12, 0.05)$. We need $P(X \geq 2) = 1 - [P(X = 0) + P(X = 1)]$. $P(X = 0) = (0.95)^{12} \approx 0.5404$. $P(X = 1) = 12(0.05)(0.95)^{11} \approx 0.3413$. Thus, $P(X \geq 2) = 1 - (0.5404 + 0.3413) = 1 - 0.8817 = 0.118$ (3 s.f.).

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Functions: Composite & Inverse

A fair coin is tossed 10 times. Find the probability of getting (a) exactly 6 heads, (b) fewer than 3 heads.

(a) $\binom{10}{6}(0.5)^{10} = 210 \times \frac{1}{1024} \approx 0.2051$. (b) $P(X \leq 2) = P(0) + P(1) + P(2) = \frac{1+10+45}{1024} = \frac{56}{1024} \approx 0.0547$.

© ClipSAT — clipsat.org | All Rights Reserved

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 9.01 Circular Measure**

[Binomial Distribution · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 1.01 Indices, Surds & Quadratics**

Find the 6th term in the expansion of $(3x - 2y^2)^8$ in fully simplified form.

In the binomial theorem expansion $\sum_{k=0}^{n} \binom{n}{k} A^{n-k} B^k$, the 1st term has $k = 0$, so the 6th term has $k = 5$. Identify terms: $n = 8$, $A = 3x$, $B = -2y^2$: $T_6 = \binom{8}{5}(3x)^{8-5}(-2y^2)^5 = \binom{8}{3}(3x)^3(-2y^2)^5$. Compute each factor:$\binom{8}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56(3x)^3 = 27x^3(-2y^2)^5 = -32y^{10}$Multiply the coefficients:

$T_6 = 56 \times 27 \times (-32)x^3y^{10} = 56 \times (-864)x^3y^{10} = -48384x^3y^{10}$.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 11.01 Kinematics**

Practice 4.1: A multiple-choice examination consists of 20 independent questions, each having 4 options of which only one is correct. A student guesses randomly on all questions. Let X be the number of correct answers. (a) State the distribution of X including its parameters. (b) Find the probability that the student passes by scoring at least 10 correct answers. (c) Calculate the mean and standard deviation of X.

(a) $X \sim B(n, p)$ with $n = 20$ and $p = 0.25$. (b) $P(X \geq 10) = 1 - P(X \leq 9)$. Using GDC `binomcdf(20, 0.25, 9)`: $P(X \leq 9) \approx 0.9861$. $P(X \geq 10) = 1 - 0.9861 = 0.0139$ (1.39%). (c) Mean: $\mu = np = 20 \times 0.25 = 5$. Standard deviation: $\sigma = \sqrt{np(1-p)} = \sqrt{20 \times 0.25 \times 0.75} = \sqrt{3.75} \approx 1.936 \approx 1.94$.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 12.01 Forces & Equilibrium**

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term a_1, and compute the exact value of the partial sum S_4.

Use the infinite sum formula $S_\infty = \frac{a_1}{1-r}$: $25 = \frac{a_1}{1-(-0.4)} = \frac{a_1}{1.4} \Rightarrow a_1 = 25 \times 1.4 = 35$. Compute the 4th partial sum $S_4 = \frac{a_1(1-r^4)}{1-r}$: $(-0.4)^4 = 0.0256$. $S_4 = \frac{35(1-0.0256)}{1.4} = \frac{35(0.9744)}{1.4} = 25 \times 0.9744 = 24.36$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An arithmetic progression has first term $a = 7$ and common difference $d = 4$. Find the 20th term u_{20}.

a. **83** ✓ **b.** 87

c. 79 **d.** 80

$u_n = a + (n-1)d \Rightarrow u_{20} = 7 + 19(4) = 7 + 76 = 83$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the sum of the first 25 terms of the AP: $3, 8, 13, 18, \ldots$

a. **1575** ✓ **b.** 1625

c. 1500 **d.** 1550

$a = 3, d = 5$. $S_{25} = \frac{25}{2}[2(3) + 24(5)] = \frac{25}{2}[6 + 120] = \frac{25}{2}(126) = 25 \times 63 = 1575$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In an AP, the 5th term is 18 and the 12th term is 39. Find the first term a and common difference d.

a. $a = 6, d = 3$ ✓ **b.** $a = 3, d = 3$

c. $a = 9, d = 3$ **d.** $a = 6, d = 4$

$a + 4d = 18$ and $a + 11d = 39$. Subtracting: $7d = 21 \Rightarrow d = 3$. Then $a = 18 - 4(3) = 6$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A geometric progression has first term $a = 5$ and common ratio $r = 2$. Find the 8th term u_8.

a. **640** ✓ **b.** 1280

c. 320 **d.** 512

$u_n = ar^{n-1} \Rightarrow u_8 = 5(2)^7 = 5(128) = 640$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the sum of the first 6 terms of the GP: $3, -6, 12, -24, \ldots$

a. -63 ✓ **b.** 63

c. -127 **d.** -65

$a = 3, r = -2$. $S_6 = \frac{3(1-(-2)^6)}{1-(-2)} = \frac{3(1-64)}{3} = 1 - 64 = -63$.

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The 2nd term of a GP is 12 and the 5th term is 324. Find the common ratio r.

a. 3 ✓

b. 2

c. 4

d. $\sqrt{3}$

$\frac{u_5}{u_2} = \frac{ar^4}{ar} = r^3 = \frac{324}{12} = 27 \Rightarrow r = 3$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

State the condition for the sum to infinity of a geometric progression to exist.

a. $|r| < 1$ ✓

b. $r > 1$

c. $r \leq 1$

d. $|r| > 1$

The infinite geometric series converges if and only if the absolute value of the common ratio is strictly less than 1: $|r| < 1$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the sum to infinity of the GP: $16, 12, 9, 6.75, \ldots$

a. **64** ✓

b. 48

c. 32

d. 72

$a = 16, r = 12/16 = 3/4$. Since $|3/4| < 1$, $S_\infty = \frac{a}{1-r} = \frac{16}{1-3/4} = \frac{16}{1/4} = 64$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Express the recurring decimal $0.\overline{27} = 0.272727\ldots$ as an exact fraction in lowest terms using an infinite geometric series.

a. $\frac{3}{11}$ ✓

b. $\frac{27}{100}$

c. $\frac{9}{33}$

d. $\frac{7}{25}$

$0.27 + 0.0027 + \ldots$ has $a = 0.27 = 27/100$ and $r = 1/100$. $S_\infty = \frac{27/100}{1-1/100} = \frac{27/100}{99/100} = \frac{27}{99} = \frac{3}{11}$.

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the coefficient of x^3 in the expansion of $(2+3x)^5$.

a. 1080 ✓

b. 720

c. 540

d. 1440

General term is $\binom{5}{r}2^{5-r}(3x)^r$. For x^3, $r=3$: $\binom{5}{3}2^2(3)^3 = 10\times 4\times 27 = 1080$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the term independent of x in the expansion of $\left(x^2+\frac{2}{x}\right)^6$.

a. **240** ✓

b. 60

c. 160

d. 120

General term is $\binom{6}{r}(x^2)^{6-r}\left(\frac{2}{x}\right)^r = \binom{6}{r}2^r x^{12-2r-r} = \binom{6}{r}2^r x^{12-3r}$. Independent when $12-3r=0 \Rightarrow r=4$. Term is $\binom{6}{4}2^4 = 15\times 16 = 240$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the expansion of $(1+ax)^6$, the coefficient of x^2 is 60. Given $a>0$, find a.

a. **2** ✓

b. 4

c. 3

d. 5

Coefficient of x^2 is $\binom{6}{2}a^2 = 15a^2$. $15a^2=60 \Rightarrow a^2=4 \Rightarrow a=2$ (since $a>0$).

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A geometric series has first term $a=-10$ and common ratio $r=1/2$. Find S_6, the sum of the first 6 terms.

a. **-315/16** ✓

b. -155/8

c. -475/16

d. -635/32

$S_6 = \frac{a(1-r^n)}{1-r} = \frac{-10(1-(1/2)^6)}{1-(1/2)} = -315/16$

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the sum of the first 40 terms of an arithmetic sequence with first term -11 and common difference -2.

a. -2000 ✓

b. -2004

c. -1994

d. -2003

$S_{40} = \frac{n}{2}(2a + (n-1)d) = \frac{40}{2}(2(-11) + (40-1)(-2)) = -2000$

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

An arithmetic sequence has first term $a = 7$ and common difference $d = 3$. Find the 7th term.

a. 21

b. 25 ✓

c. 24

d. 26

Applying Series & the Binomial Theorem principles, $u_7 = a + (n-1)d = 7 + (7-1)(3) = 25$ Therefore, the verified correct option is 25.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The 4th term of an arithmetic sequence is 19, and the common difference is 3. Find the first term.

a. 8

b. 9

c. 15

d. 10 ✓

Applying Series & the Binomial Theorem principles, $a = u_n - (n-1)d = 19 - (4-1)(3) = 10$ Therefore, the verified correct option is 10.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An arithmetic sequence has first term 6 and common difference 5. Find n such that $S_n = 1403$.

a. 19

b. 25

c. 27

d. 23 ✓

Using $S_n = \frac{n}{2}(2a + (n-1)d)$ with $S_{23} = 1403$, solving gives $n = 23$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Indices, Surds & Quadratics

A geometric series has first term $a = 11$ and common ratio $r = 1/2$. Find S_∞.

a. 11/2

b. **22** ✓

c. 22/3

d. 33

Since $|r| < 1$, $S_\infty = \frac{a}{1-r} = \frac{11}{1-(1/2)} = 22$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Kinematics

The 12th term of an arithmetic sequence is 58, and the first term is 14. Find the common difference.

a. **4** ✓

b. 1

c. 6

d. -1

$d = \frac{u_n - a}{n-1} = \frac{58-(14)}{12-1} = 4$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 12.01 Forces & Equilibrium

A geometric sequence has first term $a = 8$ and common ratio $r = 1/3$. Find the 4th term.

a. -8/27

b. **8/27** ✓

c. 8/81

d. 8/9

$u_4 = ar^{n-1} = 8(1/3)^3 = 8/27$

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 13.01 Newton's Laws

A factory produces light bulbs. The probability that a randomly selected light bulb is defective is p. The number of defective light bulbs in a random sample of n bulbs follows a binomial distribution $X \sim B(n, p)$. It is known that the mean number of defective bulbs in a sample of $n = 25$ is 2. \\ (i) Calculate the value of p. \\ (ii) For a sample of $n = 50$ bulbs, calculate the probability of finding exactly k defective bulbs, where k is the smallest integer such that $P(X \leq k) > 0.9$. \\ (iii) If a sample of $n = 50$ bulbs is taken, find the smallest value of n' such that the probability of observing at least one defective bulb is greater than 0.999 .

a. 0.04

b. **0.08** ✓

c. 0.12

d. 0.16

Part (i): For a binomial distribution $X \sim B(n, p)$, the mean is given by $E(X) = np$. \\ We are given $n = 25$ and $E(X) = 2$. \\ So, $25p = 2$, which gives $p = \frac{2}{25} = 0.08$. \\ Part (ii): Now, $n = 50$ and $p = 0.08$. The number of defective bulbs follows $X \sim B(50, 0.08)$. \\ We need to find the smallest integer k such that $P(X \leq k) > 0.9$. \\ We can use a binomial cumulative distribution table or a calculator. \\ Let's calculate a few values: $P(X \leq 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$ Using.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 14.01 Energy, Work & Power

A fair six-sided die is rolled 4 times. Let X be the number of times a 6 is obtained. The possible values for X are 0, 1, 2, 3, and 4. The probability of obtaining a 6 on any given roll is $p = \frac{1}{6}$. The number of trials is $n = 4$. This scenario can be modelled by a binomial distribution, denoted as $X \sim B(n, p)$. Which of the following represents the correct parameters for this binomial distribution?

a. $B(4, 1/6)$ ✓

b. $B(6, 1/4)$

c. $B(1, 4/6)$

d. $B(4, 4/6)$

The problem describes a situation with a fixed number of independent trials (rolling a die 4 times), where each trial has only two possible outcomes (obtaining a 6 or not obtaining a 6), and the probability of success (obtaining a 6) is constant for each trial. In a binomial distribution, denoted as $X \sim B(n, p)$, n represents the number of trials, and p represents the probability of success on a single trial. From the question: - The number of times the die is rolled is 4, so $n = 4$. - The probability of o.

© ClipSAT — clipsat.org | All Rights Reserved

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 15.01 Representation of Data

A die is rolled 15 times; X is the number of sixes. Is X binomially distributed?

a. No — trials are not independent / probability is not constant

b. Yes — but only if $n > 30$

c. **Yes — fixed, independent trials with constant success probability ✓**

d. No — the number of trials is not fixed

The trials are fixed in number, independent, and each has the same probability of success, so the conditions for a binomial distribution are satisfied.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 16.01 Permutations & Combinations

A fair coin is tossed 20 times; X is the number of heads. Is X binomially distributed?

a. No — trials are not independent / probability is not constant

b. Yes — but only if $n > 30$

c. **Yes — fixed, independent trials with constant success probability ✓**

d. No — the number of trials is not fixed

The trials are fixed in number, independent, and each has the same probability of success, so the conditions for a binomial distribution are satisfied.

© ClipSAT — clipsat.org | All Rights Reserved

45 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 17.01 Discrete Random Variables**

Cards are dealt from a single shuffled deck without replacement; X is the number of aces dealt. Is X binomially distributed?

a. **No — trials are not independent / probability is not constant ✓**

b. Yes — but only if $n > 30$

c. Yes — fixed, independent trials with constant success probability

d. No — the number of trials is not fixed

The conditions for a binomial distribution (fixed independent trials with constant probability) are not satisfied here.

© ClipSAT — clipsat.org | All Rights Reserved